When does data quality help human deferral? Knowledge-based systems for AI governance and managerial decision-making

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Short communication.

Abstract

AI governance must allocate limited review capacity while resolving cases that quality rules bar from automation. We study a knowledge-based router combining calibrated predictions, quality rules, loss estimation from incomplete logs and capacity-constrained allocation. Sorting solves the stated allocation in feasible and overloaded regimes; with fixed mandatory demand, review ordering does not change backlog. We compare learned routers with cost-aware heuristics on four credit datasets, nine quality conditions, five simulated reviewers and 20 repeated splits. No learned router beat Chow's cost-risk rule when pooled. A cost-sensitive one-vs-all surrogate and the two-head estimator were the strongest learned routers, without a significant difference. Both beat Chow on all 20 splits with about 6,000 gain-fitting records (Holm $p < 0.001$) but lost with a few hundred; the two-head estimator was more data-efficient at smaller volumes. Adding quality inputs to the automation-loss head reduced cost under informative missingness on 20 splits, but this exploratory effect did not survive Holm adjustment. A 40-day simulation showed that staffing determined backlog and cohort service levels more than prioritization. These results support evaluating estimation, rule coverage and review capacity separately.

Keywords: Knowledge-based systems; Learning to defer; Data quality; AI governance; Human–AI collaboration; Review capacity; Managerial decision-making

1 Introduction

Organizations route credit, claims and eligibility decisions between models and reviewers. Governance links this division of labor to accountability and data controls [1, 2, 3, 4], but oversight alone does not establish effective control [5]. Managers must decide which records permit automation, where review helps and whether staffing meets compulsory demand. Quality-triggered rules add mandatory work during deterioration; unserved cases remain unresolved.

Existing methods address parts of this problem. Rejection and learning to defer (L2D) trade model error against abstention or expert error [6, 7, 8, 9]; rejection wrappers and uncertainty quantification provide confidence-based signals [10, 11]; triage estimates human and model error separately [12]; and recent work adds workload limits [13],

sparse expert data [14], quality-aware cascades [15] and congestion [16]. Data-management research measures completeness and validity [17, 18, 19], human-in-the-loop research studies how human effort is deployed [20, 21], and knowledge-based credit decision support links predictions to business actions [22]. We examine quality-triggered mandatory review, capacity shortfalls and unresolved cases together, distinguishing whether quality predicts model error, reviewer error, both or neither.

We ask: *under what data-quality and review-capacity conditions does quality-aware deferral reduce decision cost while satisfying explicit governance rules?* The study differs from L2D work in two structural ways: rules make some records non-automatable and some unresolvable, which changes the optimal allocation; and the allocator is separated from the loss estimator, so published surrogates and our estimator compete inside the same governance layer. Contributions: (i) an exact solution of governance-constrained allocation in both regimes, correcting the earlier overload rule, and a proof that backlog is invariant to prioritization; (ii) a matched comparison of learned routers against heuristics under diverse simulated reviewers, defects invisible to quality checks and an external dataset; and (iii) a conditions analysis with censoring-aware service measures for managers. We claim neither a new surrogate loss nor a uniformly better router.

2 Method

2.1 Governance-constrained allocation

For record i in a batch of n, a calibrated predictor gives p_i; with costs $c_{FN} = 5$, $c_{FP} = 1$, the automated decision is $\hat{y}_i = \mathbf{1}[p_i \geq 1/6]$ with loss $\ell_i^A = w(y_i)\mathbf{1}[\hat{y}_i \neq y_i]$, $w(1) = 5$, $w(0) = 1$. A reviewer errs with probability e_i, giving expected loss $\ell_i^H = w(y_i)e_i$; review costs $c_R = 0.05$ and an unresolved (held) record $c_H = 1.5$. Observable quality $q_i = (u_i, v_i)$ is the missing-field and out-of-range fraction, with bounds $[Q_{.01,j} - 0.1t_j,\ Q_{.99,j} + 0.1t_j]$, $t_j = \max(Q_{.99,j} - Q_{.01,j}, 1)$, from predictor-training rows. The knowledge base marks record i mandatory, $m_i = \mathbf{1}[u_i \geq 0.5 \vee v_i \geq 0.25]$; thresholds are illustrative organizational policy. With actions $a_i, r_i, h_i \in \{0, 1\}$ (automate, review, hold) and capacity $B = \lfloor bn \rfloor$, the router solves

$$\min \tfrac{1}{n} \textstyle\sum_i \left[a_i \ell_i^A + r_i(\ell_i^H + c_R) + h_i c_H \right] \tag{1}$$

subject to $a_i + r_i + h_i = 1$, $\sum_i r_i \leq B$, $a_i \leq 1 - m_i$, $h_i \leq m_i$ and the service-first constraint $\sum_i h_i = (|M| - B)^+$, $M = \{i : m_i = 1\}$. Let $g_i = \ell_i^A - \ell_i^H - c_R$.

Proposition 1. *If $|M| \leq B$, reviewing all of M plus up to $B - |M|$ optional records with the largest positive g_i is optimal. If $|M| > B$, every feasible route reviews exactly B mandatory records and no optional record; reviewing the B smallest ℓ_i^H is optimal. Routes do not depend on c_H.* Proof: in the first case no record is held, so each

optional review changes (1) by $-g_i/n$ and an exchange argument applies. In the second, B unheld mandatory records must be reviewed, exhausting capacity; only $\sum_{i\in R}\ell_i^H$ depends on the reviewed set R, and c_H enters as the constant $(|M|-B)c_H$. □ Gain ordering under overload wrongly includes automated loss for records that cannot be automated.

Proposition 2. *Assume fixed mandatory flags, identical arrivals, unit-time reviews and no abandonment. If held records carry over and Proposition 1 is applied each period, holds, backlog, reviews and completions in every period are identical for any ordering of mandatory records.* Proof: by induction, equal backlog implies equal $|M_t|$, hence equal holds $(|M_t|-B)^+$. □ Staffing therefore sets the backlog, approximately growing by $(\mu_t-b)^+n$ per period for mandatory rate μ_t, whereas prioritization only decides who waits.

2.2 Loss estimation

The allocator accepts any estimates of ℓ_i^A and ℓ_i^H. Our two-head estimator fits two gradient-boosted regressors on the eight inputs $z_i=(p_i, s_i, q_i, p_iq_i, \hat{y}_iq_i)$, where $s_i=\min(5p_i, 1-p_i)$ is the Bayes risk of automation: an *automation head* $\hat{\ell}^A(z)$ fitted on all labeled records, and a *reviewer head* $\hat{\ell}^H(z)$ fitted only on records with one logged review decision, using the target, unbiased under randomized logging, $w(y_i)\mathbf{1}[\text{review}\neq y_i]$. Optional records are reviewed only if $\hat{g}_i=\hat{\ell}^A-\hat{\ell}^H-c_R>0$; overloaded mandatory records are ordered by $\hat{\ell}^H$. Allocation costs $O(n\log n)$; each decision is logged as a label-free audit tuple $(p_i, q_i, m_i, \hat{\ell}_i^A, \hat{\ell}_i^H, \text{action})$, which documents routing rather than explaining the model [23, 24, 25].

3 Experimental protocol

We use the credit benchmarks [26] German (1,000 records), Australian (690) and default (30,000) [27, 28, 29], and HMEQ (5,960; natural missingness in 11 of 12 predictors) as an external dataset unused during development [30]. Feature-identical records are grouped and split by majority label into 45% predictor training, 15% calibration, 20% gain fitting and 20% testing over seeds 2–21. A histogram gradient-boosting classifier (at most 120 iterations, 15 leaves) is calibrated by logistic regression; loss heads are boosted regressors (at most 100 iterations, seven leaves); calibration and gain partitions are expanded into clean, missing, outlier and mixed copies.

Test sets face nine conditions: clean; Family A, defects visible to quality checks (missing, outliers, mixed, and a held-out severe shift with 35–75% missing cells); and Family B, domain-valid defects never used in fitting (outcome-informative missingness, stale records with continuous fields deflated by 10–45%, inconsistent but valid category codes, and a realistic mix). Five reviewer models are simulated: constant ($e=0.10$), quality-coupled ($\min(0.4, 0.05+0.35u+0.25v)$, the earlier design), feature-dependent (a hidden projection of clean features), AI-

correlated (errors concentrated where the model errs) and adverse (0.25); only the second uses the router's signals. Learned methods train on noisy, incomplete logs: a random 40% of expanded gain-fitting rows carry one sampled reviewer decision.

All methods share the predictor, rules, costs, capacity and allocator. Comparators are confidence ($-|p_i - 1/6|$), Chow cost risk (s_i) [6], quality only ($u_i + v_i$), the earlier single-head gain, L2D-CE with cost-sensitive weights including the review fee [8], L2D-OvA [9] trained as published and routed by a cost-sensitive plug-in rule on its probability estimates (expected automation loss minus expected review loss minus c_R), differential triage [12] and a probability-only two-head ablation; L2D models are one-hidden-layer rejectors. Their deferral scores also rank mandatory cases under overload; two-head routing instead uses estimated reviewer loss. Heuristics fill capacity; learned methods review only positive scores. As splits reuse observations, tests use the corrected resampled t-test [31] with Holm adjustment [32] within declared families: pooled comparisons (8 tests), per-dataset comparisons (9), data-volume comparisons (7) and Table 2 (18, exploratory). A 40-day simulation (100 arrivals per day, severe shift on days 11–20, held cases carried forward at 0.25 per case-day, c_H for cases open at day 40) measures service among mandatory arrivals with enough follow-up for each target; eligible unresolved cases are unmet. Resolution-time quantiles use Kaplan–Meier with censoring [33].

4 Results and discussion

Heuristics lead in pooled data; learned routers need data. Constrained methods reviewed about 19.7–19.8% and held 6.0% of records with zero violations; costs reflect selection and optional admission (Table 1). Pooled, Chow (0.5291) and confidence (0.5295) were cheapest; the best learned routers, L2D-OvA (0.5312) and the two-head estimator (0.5355), did not differ significantly (14/20 splits favored OvA; $p = 0.48$). The two-head router beat L2D-CE by 1.95% (19/20; Holm $p = 0.11$) and trailed Chow by 1.2% (5/20; Holm $p = 0.70$). Chow or confidence was cheapest under four of five reviewer models, OvA under the feature-dependent one. Dataset size explains the pattern: with about 6,000 gain-fitting records (default data), OvA and the two-head router beat Chow by 1.4% and 1.1% on all 20 splits (Holm $p < 0.001$); with 140–200 records (German, Australian) both lost. Subsampling the default partition (Fig. 1) left the two-head router 2.0% costlier than Chow at about 200 records, approximately competitive at 1,800 (+0.1%, 10/20, $p = 0.85$) and cheaper only at 6,000 (-1.1%, 20/20). It was the most data-efficient learned router, beating OvA on all 20 splits at 200 and 600 records (by 3.0% and 2.7%; Holm $p < 0.005$), whereas OvA was cheapest at 6,000 (-1.4%). This shows data dependence in one dataset, not a deployment threshold.

Quality signals act on automated error (exploratory). Adding quality inputs to one head at a time (Table 2) shows that under outcome-informative missingness they reduced cost through the automation head (2.8%, 20/20

Table 1: Expected cost per record at 20% capacity (lower is better), averaged over eight non-clean conditions, five reviewer models and 20 splits; HMEQ is evaluated on its natural data. Learned methods are trained on noisy, incomplete review logs. Bold: lowest deployable cost per column.

Method	German	Australian	Default	Macro	HMEQ (natural)
Automation only (rules ignored)[a]	0.6254	0.4046	0.6291	0.5530	0.2583
Random	0.6327	0.4752	0.6012	0.5697	0.2700
Confidence	0.6045	**0.4139**	0.5700	0.5295	**0.1847**
Chow cost risk [6]	**0.5987**	0.4204	0.5684	**0.5291**	0.2288
Quality only	0.6514	0.4807	0.5901	0.5741	0.2960
Single-head gain (earlier version)	0.6123	0.4329	0.5634	0.5362	0.2065
L2D-CE [8]	0.6219	0.4469	0.5695	0.5461	0.2182
L2D-OvA [9]	0.6086	0.4247	**0.5604**	0.5312	0.2103
Differential triage [12]	0.6108	0.4352	0.5657	0.5372	0.2082
Two-head, probability only	0.6100	0.4338	0.5705	0.5381	0.2163
Two-head, quality-aware (proposed)	0.6104	0.4338	0.5622	0.5355	0.2053
Oracle (uses labels)[b]	0.4724	0.2904	0.3692	0.3773	0.0624

[a]Ignores the rules (12.6% of records automated in violation). [b]Uses true labels and reviewer losses; an optimistic lower bound on attainable cost, not deployable.

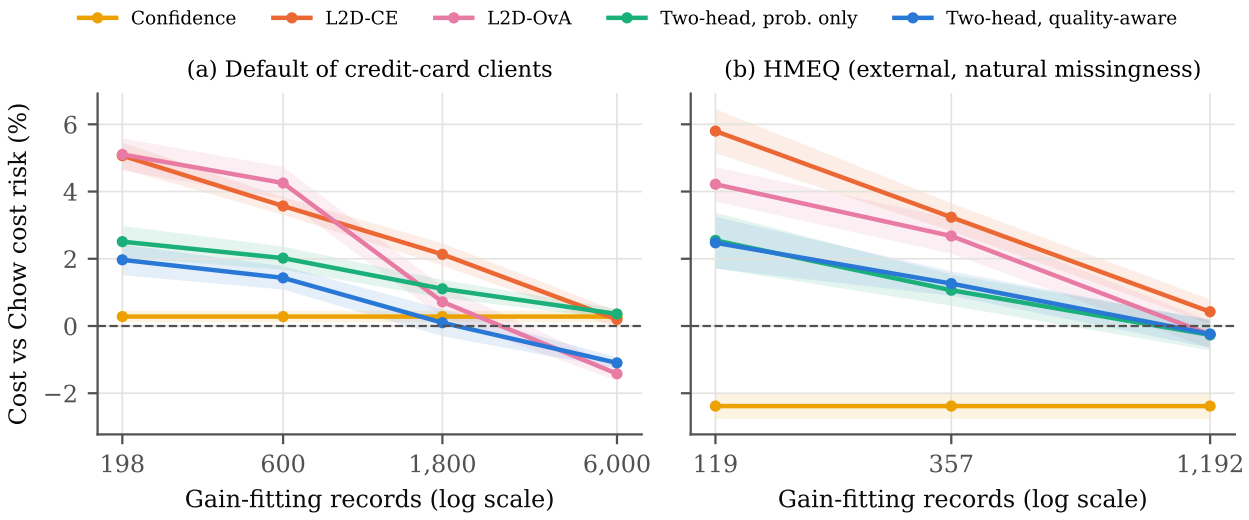


Figure 1: Learned routing versus data volume on (a) default and (b) HMEQ data. Cost relative to Chow's rule as the gain-fitting partition grows (20% capacity, 20 splits per size); negative values favor the learned router. Bands: ±1.96 standard errors describe split variability, not population confidence. Generated from saved outputs using AI-assisted analysis code.

splits) but not the reviewer head; HMEQ's natural missingness showed the same pattern (+4.5%, 17/20, versus +1.0%). This is consistent with incompleteness predicting the model's own error, but no cell survives Holm adjustment, so the mechanism awaits confirmatory testing. Through the reviewer head, quality inputs helped only when reviewer error was built from quality (+0.7%, 16/20), so the earlier simulation favored the representation slightly. Effects for stale records and miscoded categories, invisible to quality checks, were small. Ordering overloaded records by $\hat{\ell}^H$ rather than gain lowered cost by only 0.18% ($p = 0.47$), and two heads beat a single head only directionally (+0.6% and +0.5% with 10% and 40% of rows logged, −0.5% with complete logs; all $p > 0.5$).

Table 2: Value of quality inputs by estimator head: cost reduction (%) relative to the probability-only two-head router at 20% capacity, with splits won out of 20 in parentheses.

Condition	Automation head only	Reviewer head only	Both heads (proposed)
Informative missingness (B)	+2.83 (20)†	−0.28 (7)	+2.29 (18)†
Stale records (B)	+0.98 (13)	+0.04 (10)	+0.91 (14)
Inconsistent codes (B)	+0.15 (10)	−0.23 (7)	−0.08 (10)
Visible defects (A), quality-coupled reviewer	+0.07 (13)	+0.70 (16)	+0.65 (16)
Visible defects (A), four decoupled reviewers	+0.18 (11)	−0.24 (5)	−0.07 (10)
HMEQ, natural missingness (external)	+4.53 (17)†	+0.95 (14)	+5.12 (19)†

Bold: Holm-adjusted $p < 0.05$ over the 18 cells (none). †Unadjusted corrected $p < 0.05$ only. Exploratory analysis.

Rules protect only against visible defects. Under severe shift the rules forced 47.5% of records into holds and raised cost by 67% relative to an unconstrained router, which automated 54% of policy-barred records. Stricter thresholds cut the share of truly defective records automated from 80% to 4.5% under visible shift but left 75% automated under invisible defects, so zero violations do not establish adequate control. Re-optimizing every method for cost ratios of 2, 5 and 10, review fees of 0–0.2 and hold penalties of 0.5–3.0 kept confidence cheapest in all seven settings, with the best learned router within 0.7% of Chow; hold penalties shifted cost linearly with identical routes, as Proposition 1 predicts.

Staffing, not routing, sets service levels. Mandatory demand rose from 5.5% to 70% of arrivals during the simulated episode (Fig. 2). As Proposition 2 predicts, all prioritization rules produced identical backlogs, and the fluid estimate matched peaks of 500, 300 and 200 records at 20%, 40% and 50% capacity. Under cost ordering at 20% capacity, 21.7% of mandatory cases remained unresolved at day 40. Two-day resolution was 36% among eligible arrivals versus 48% among resolved cases; the 95th-percentile resolution time was not estimable in 72 of 80 runs. At 40% capacity all cases were resolved and cost per arrival fell by 45%. Prioritization only redistributed waiting: at 40% capacity first-in-first-out cut the Kaplan–Meier 95th-percentile resolution time from 13.9 to 7.1 days

and raised seven-day resolution from 76% to 94%, but lowered two-day resolution from 65% to 40%. Managers should therefore validate any learned router against a cost-aware heuristic on local data, staff to mandatory rather than average demand, choose the queue discipline for its service profile, and report cohort service levels and defect coverage, not violations alone [34, 35].

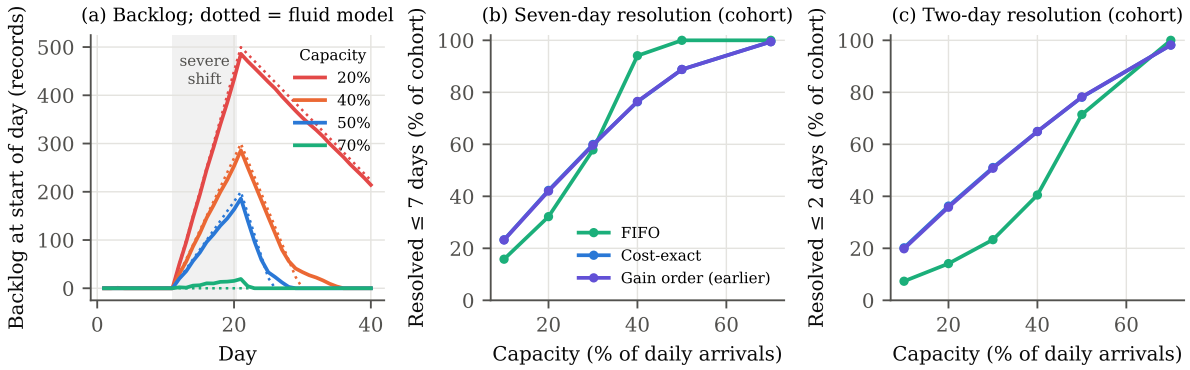


Figure 2: Backlog and cohort service under a severe-shift episode (days 11–20). (a) Backlog at four staffing levels; dotted lines are the fluid approximation. (b) Seven-day and (c) two-day resolution among eligible mandatory arrivals, counting unresolved eligible cases as unmet, by prioritization rule (four datasets, two reviewers, ten seeds). Generated from saved outputs using AI-assisted analysis code.

5 Limitations

Reviewers are simulated; real reviewers may anchor on model output or tire under backlog, so validation with review logs is the main next step. Benchmarks are old, Family B defects stylized, gradual drift unmodeled [36], and thresholds and costs illustrative. Quality effects remain exploratory. OvA approximates reviewer loss by multiplying conditional mean error and class cost, assuming conditional independence; this can fail for correlated errors. Baselines are adapted rather than tuned, logging is random, and confirmatory data-volume evidence, fairness and legal sufficiency remain untested.

6 Conclusion

Separating rules, loss estimation and capacity makes governance-constrained routing exact and auditable for the stated loss estimates. No learned router beat a cost-aware heuristic in pooled data; learned routers helped where review history was large, and quality signals appeared useful under informative missingness pending confirmation. Under deterioration, staffing, not routing, determined service completion.

Data and code availability Experimental code and outputs: public repository, commit `03166e9`. Datasets: [27, 28, 29, 30]. [Author to confirm the final data-deposit statement and any archival release.]

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work the author used ChatGPT (OpenAI) for original study design, code and drafting, and final editorial and verification assistance, and Claude (Anthropic) to reimplement and extend the experiments, implement baselines, analyze results, generate the two data figures with analysis code, check literature and draft this revision. [After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.]

References

[1] R. Abraham, J. Schneider, J. vom Brocke, Data governance: A conceptual framework, structured review, and research agenda, Int. J. Inf. Manag. 49 (2019) 424–438. doi:10.1016/j.ijinfomgt.2019.07.008.

[2] M. Janssen, P. Brous, E. Estevez, L.S. Barbosa, T. Janowski, Data governance: Organizing data for trustworthy artificial intelligence, Gov. Inf. Q. 37 (2020) 101493. doi:10.1016/j.giq.2020.101493.

[3] E. Papagiannidis, P. Mikalef, K. Conboy, Responsible artificial intelligence governance: A review and research framework, J. Strateg. Inf. Syst. 34 (2025) 101885. doi:10.1016/j.jsis.2024.101885.

[4] B.W. Wirtz, J.C. Weyerer, I. Kehl, Governance of artificial intelligence: A risk and guideline-based integrative framework, Gov. Inf. Q. 39 (2022) 101685. doi:10.1016/j.giq.2022.101685.

[5] B. Green, The flaws of policies requiring human oversight of government algorithms, Comput. Law Secur. Rev. 45 (2022) 105681. doi:10.1016/j.clsr.2022.105681.

[6] C. Chow, On optimum recognition error and reject tradeoff, IEEE Trans. Inf. Theory 16 (1970) 41–46. doi:10.1109/TIT.1970.1054406.

[7] X.Y. Zhang, G.S. Xie, X. Li, T. Mei, C.L. Liu, A survey on learning to reject, Proc. IEEE 111 (2023) 185–215. doi:10.1109/JPROC.2023.3238024.

[8] H. Mozannar, D. Sontag, Consistent estimators for learning to defer to an expert, in: Proc. 37th Int. Conf. Mach. Learn., PMLR 119 (2020) 7076–7087.

[9] R. Verma, E. Nalisnick, Calibrated learning to defer with one-vs-all classifiers, in: Proc. 39th Int. Conf. Mach. Learn., PMLR 162 (2022) 22184–22202.

[10] J. Mena, O. Pujol, J. Vitrià, Uncertainty-based rejection wrappers for black-box classifiers, IEEE Access 8 (2020) 101721–101746. doi:10.1109/ACCESS.2020.2996495.

[11] M. Abdar, F. Pourpanah, S. Hussain, et al., A review of uncertainty quantification in deep learning: Techniques, applications and challenges, Inf. Fusion 76 (2021) 243–297. doi:10.1016/j.inffus.2021.05.008.

[12] M. Raghu, K. Blumer, G. Corrado, J. Kleinberg, Z. Obermeyer, S. Mullainathan, The algorithmic automation problem: Prediction, triage, and human effort, arXiv preprint (2019). https://doi.org/10.48550/arXiv.1903.12220.

[13] J.V. Alves, D. Leitão, S. Jesus, M.O.P. Sampaio, J. Liébana, P. Saleiro, M.A.T. Figueiredo, P. Bizarro, Cost-sensitive learning to defer to multiple experts with workload constraints, Trans. Mach. Learn. Res. (2024). https://openreview.net/forum?id=TAvGZm2Rqb.

[14] P. Hemmer, L. Thede, M. Vössing, J. Jakubik, N. Kühl, Learning to defer with limited expert predictions, Proc. AAAI Conf. Artif. Intell. 37 (2023) 6002–6011. doi:10.1609/aaai.v37i5.25742.

[15] A. Farinhas, N.M. Guerreiro, S. Agrawal, R. Rei, A. Martins, Translate smart, not hard: Cascaded translation systems with quality-aware deferral, in: Proc. 2025 Conf. Empirical Methods Nat. Lang. Process. (2025) 26742–26756. doi:10.18653/v1/2025.emnlp-main.1358.

[16] V. Saketos, W. Reveillard, A. Proutiere, S. Dalleiger, M. Xiao, To defer or not to defer? Online learning with LLMs and congested experts, arXiv preprint (2026). https://doi.org/10.48550/arXiv.2610.15128.

[17] C. Chai, J. Wang, Y. Luo, Z. Niu, G. Li, Data management for machine learning: A survey, IEEE Trans. Knowl. Data Eng. 35 (2023) 4646–4667. doi:10.1109/TKDE.2022.3148237.

[18] L. Zhang, D. Jeong, S. Lee, Data quality management in the Internet of Things, Sensors 21 (2021) 5834. doi:10.3390/s21175834.

[19] S. Mohammed, L. Budach, M. Feuerpfeil, N. Ihde, A. Nathansen, N. Noack, H. Patzlaff, F. Naumann, H. Harmouch, The effects of data quality on machine learning performance on tabular data, Inf. Syst. 132 (2025) 102549. doi:10.1016/j.is.2025.102549.

[20] X. Wu, L. Xiao, Y. Sun, J. Zhang, T. Ma, L. He, A survey of human-in-the-loop for machine learning, Future Gener. Comput. Syst. 135 (2022) 364–381. doi:10.1016/j.future.2022.05.014.

[21] S. Tariq, M. Baruwal Chhetri, S. Nepal, C. Paris, A^2C: A modular multi-stage collaborative decision framework for human–AI teams, Expert Syst. Appl. 282 (2025) 127318. doi:10.1016/j.eswa.2025.127318.

[22] M. Abbaspour Onari, M. Jahangoshai Rezaee, M. Saberi, M.S. Nobile, An explainable data-driven decision support framework for strategic customer development, Knowl.-Based Syst. 295 (2024) 111761. doi:10.1016/j.knosys.2024.111761.

[23] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, et al., Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI, Inf. Fusion 58 (2020) 82–115. doi:10.1016/j.inffus.2019.12.012.

[24] A. Adadi, M. Berrada, Peeking inside the black-box: A survey on explainable artificial intelligence (XAI), IEEE Access 6 (2018) 52138–52160. doi:10.1109/ACCESS.2018.2870052.

[25] A. Artelt, R. Visser, B. Hammer, "I do not know! but why?" — Local model-agnostic example-based explanations of reject, Neurocomputing 558 (2023) 126722. doi:10.1016/j.neucom.2023.126722.

[26] S. Lessmann, B. Baesens, H.V. Seow, L.C. Thomas, Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research, Eur. J. Oper. Res. 247 (2015) 124–136. doi:10.1016/j.ejor.2015.05.030.

[27] [dataset] H. Hofmann, Statlog (German Credit Data), UCI Machine Learning Repository (1994). https://doi.org/10.24432/C5NC77.

[28] [dataset] R. Quinlan, Statlog (Australian Credit Approval), UCI Machine Learning Repository (1987). https://doi.org/10.24432/C59012.

[29] [dataset] I. Yeh, Default of Credit Card Clients, UCI Machine Learning Repository (2009). https://doi.org/10.24432/C55S3H.

[30] [dataset] B. Baesens, D. Rösch, H. Scheule, Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS, Wiley, Hoboken, NJ, 2016 (HMEQ data set; companion data: https://www.creditriskanalytics.net/datasets-private2.html, accessed 23 October 2026).

[31] C. Nadeau, Y. Bengio, Inference for the generalization error, Mach. Learn. 52 (2003) 239–281. doi:10.1023/A:1024068626366.

[32] S. Holm, A simple sequentially rejective multiple test procedure, Scand. J. Stat. 6 (1979) 65–70.

[33] E.L. Kaplan, P. Meier, Nonparametric estimation from incomplete observations, J. Am. Stat. Assoc. 53 (1958) 457–481. doi:10.1080/01621459.1958.10501452.

[34] M.H. Jarrahi, Artificial intelligence and the future of work: Human-AI symbiosis in organizational decision making, Bus. Horiz. 61 (2018) 577–586. doi:10.1016/j.bushor.2018.03.007.

[35] P. Mikalef, M. Gupta, Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance, Inf. Manag. 58 (2021) 103434. doi:10.1016/j.im.2021.103434.

[36] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, G. Zhang, Learning under concept drift: A review, IEEE Trans. Knowl. Data Eng. 31 (2019) 2346–2363. doi:10.1109/TKDE.2018.2876857.